

		the sum of heat supplied to the system and the work done on the system.
16	C	Maximum force = 8 N, amplitude = 0.5m $F = ma_0 = m\omega^2 x_0 \rightarrow \omega = \sqrt{8} \text{ rads}^{-1}$ maximum velocity = $\omega x_0 = \sqrt{8} \times 0.5 = 1.41 \text{ ms}^{-1}$
17	D	P